

VINGA, Sweden

WMO: 025160

Lat: 57.6322N

Lon: 11.6048E

Elev: 18

StdP: 101.11

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.1	-6.4	-12.5	1.3	-5.7	-10.3	1.6	-4.4	18.9	3.0	17.3	3.8	6.3	50	0.764

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.5	23.2	18.9	22.0	18.5	20.9	17.6	20.0	22.3	19.0	21.4	18.1	20.5	4.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	13.9	21.6	17.9	12.9	20.8	17.0	12.1	19.7	57.4	22.4	54.1	21.6	51.1	20.5	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.1	15.2	13.7	DB	-8.1	25.1	3.0	1.4	-10.3	26.1	-12.1	26.9	-13.7	27.7	-15.9	28.7	(4)
(5)			WB	-8.2	20.3	2.9	1.4	-10.3	21.3	-11.9	22.1	-13.6	22.9	-15.7	24.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	1.1	0.9	2.7	6.9	11.5	15.0	17.7	17.6	14.5	10.1	5.9	3.0	(6)	
	DBStd	6.69	3.59	3.52	2.80	2.70	3.01	2.23	2.26	1.88	2.03	2.44	3.19	3.65	(7)	
	HDD10.0	1244	276	255	227	101	15	0	0	0	1	28	125	217	(8)	
	HDD18.3	3462	534	488	485	344	212	103	40	37	117	254	373	475	(9)	
	CDD10.0	863	0	0	0	7	62	150	239	236	134	33	2	0	(10)	
	CDD18.3	41	0	0	0	0	1	4	21	14	1	0	0	0	(11)	
	CDH23.3	32	0	0	0	0	2	2	22	6	0	0	0	0	(12)	
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)	
(14) Wind	WSAvg	7.0	7.9	7.3	6.6	5.9	5.9	6.1	5.8	6.4	7.5	8.0	8.3	8.1	(14)	
(15) Precipitation	PrecAvg	562	42	27	34	32	40	45	50	60	65	61	56	49	(15)	
	PrecMax	723	95	78	70	69	105	96	126	154	176	155	106	130	(16)	
	PrecMin	433	1	0	1	0	6	3	7	6	8	21	12	11	(17)	
	PrecStd	77	25	19	20	17	21	23	30	36	34	30	26	26	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	6.9	10.4	17.9	22.5	23.2	25.5	24.2	20.5	15.4	12.7	9.2	(19)	
		MCWB	6.9	5.4	7.2	11.7	16.2	17.1	19.4	19.8	16.8	13.6	11.9	8.1	(20)	
	2%	DB	7.4	6.1	7.8	14.0	19.5	21.0	23.6	22.8	18.9	14.5	11.5	8.6	(21)	
		MCWB	6.1	5.1	5.9	9.8	14.6	16.9	19.0	19.3	15.8	13.0	10.5	7.5	(22)	
	5%	DB	6.5	5.4	6.6	12.0	17.4	19.6	22.2	21.8	18.0	13.9	10.8	7.9	(23)	
		MCWB	5.3	4.5	5.2	8.9	13.6	16.1	18.6	18.8	15.6	12.4	9.7	7.0	(24)	
	10%	DB	5.6	4.8	5.7	10.5	15.7	18.3	21.2	20.6	17.2	13.3	9.9	7.2	(25)	
		MCWB	4.6	4.0	4.6	7.9	12.6	15.3	17.9	17.8	15.0	11.8	8.8	6.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	6.0	7.7	11.7	17.5	18.6	20.7	21.4	17.7	14.3	12.0	8.7	(27)	
		MCDB	8.2	6.4	9.9	14.3	21.2	21.2	23.5	23.1	18.9	14.9	12.5	9.0	(28)	
	2%	WB	6.4	5.3	6.5	10.2	15.7	17.5	19.9	20.4	16.5	13.4	10.7	7.9	(29)	
		MCDB	7.1	5.8	7.7	12.5	18.6	19.9	22.6	22.2	17.7	14.2	11.3	8.3	(30)	
	5%	WB	5.6	4.6	5.6	9.3	14.4	16.6	19.1	19.2	15.9	12.7	9.9	7.2	(31)	
		MCDB	6.3	5.2	6.4	11.6	16.8	18.9	21.8	21.3	17.2	13.6	10.7	7.8	(32)	
	10%	WB	4.8	4.1	4.9	8.3	13.1	15.9	18.3	18.2	15.2	12.2	9.0	6.5	(33)	
		MCDB	5.6	4.7	5.6	10.2	15.2	18.1	20.9	20.4	16.8	13.2	9.9	7.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	2.8	2.6	3.0	3.6	3.9	3.5	3.5	3.2	3.0	2.7	2.5	2.7	(35)	
		MCDBR	3.0	2.3	3.8	5.9	6.6	5.3	4.9	4.5	3.7	2.5	2.4	2.5	(36)	
	5% WB	MCWBR	3.3	2.2	2.6	3.6	4.0	3.7	3.4	3.3	2.9	2.8	2.8	2.8	(37)	
		MCWBR	2.9	2.2	3.3	5.0	5.9	4.6	4.5	3.9	3.1	2.4	2.4	2.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.304	0.331	0.348	0.386	0.380	0.369	0.391	0.391	0.360	0.347	0.321	0.289	(40)	
		taud	2.254	2.314	2.353	2.299	2.396	2.454	2.410	2.406	2.477	2.413	2.302	2.208	(41)	
	Ebn at Noon		548	705	802	823	856	871	840	812	786	682	521	446	(42)	
		Edh at Noon	49	72	90	111	107	103	105	99	80	65	47	38	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.36	0.93	2.39	4.03	5.38	5.98	5.48	4.28	2.89	1.33	0.48	0.25	(44)	
	RadStd		0.07	0.17	0.23	0.47	0.44	0.42	0.52	0.29	0.28	0.18	0.08	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (59 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	-124	-177	N/A	N/A

Nomenclature: See separate page